

Comprehensive Analysis Report: Seizure Prediction with EEG Data

Preprocessing, Model Complexity, and Regularization Effects on Generalization Performance

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Abstract—This report studies the impact of preprocessing decisions, model complexity, regularization, and class-imbalance handling on seizure prediction performance across three EEG datasets. By comparing different preprocessing pipelines, imbalance strategies, and regularization techniques, the study identifies the main drivers of generalization in biomedical signal processing. The strongest gains come from preprocessing design, while L1, L2, and Elastic Net regularization show only minor performance differences.

Index Terms—EEG, seizure prediction, preprocessing, logistic regression, regularization, class imbalance, PCA, feature selection

I. OVERVIEW

This is a thorough study of the impact of preprocessing decisions, model complexity, and regularization techniques on seizure prediction performance in three different EEG datasets. By comparing different pipelines, imbalance methods, and regularization techniques, the work builds practical insight into generalization in biomedical signal processing.

Key Findings

- The order of preprocessing steps is a major factor affecting performance. Pipeline A and Pipeline B perform differently across datasets, with Pipeline B often performing better.
- The three regularization methods, L1, L2, and Elastic Net, show almost no difference in performance, suggesting preprocessing matters more than regularization choice.
- The effectiveness of class-imbalance handling differs by preprocessing pipeline. Undersampling is more effective in some reduced-dimensionality settings.
- Learning curves clearly reveal underfitting and overfitting through train-validation gaps.
- L1 feature selection becomes increasingly sparse as regularization strength changes, eventually reaching 100% sparsity at strong regularization.

II. DATASET OVERVIEW & JUSTIFICATION

These three datasets provide complementary perspectives. Dataset 1 is highly imbalanced and very high-dimensional, Dataset 2 is moderately imbalanced and moderately dimensional, and Dataset 3 is balanced but

TABLE I
DATASET 1: EEG_SCALED_DATA.CSV

Property	Value	Description	Relevance
Samples	11,233	Total EEG records	Large enough for training
Features	36,864	Raw EEG channels (multi-electrode)	High-dimensional; requires dimensionality reduction
Class Balance	Imbalanced	Seizure vs non-seizure, about 8.6% minority class	Tests imbalance handling
Feature Type	Raw signals	Time-series EEG values	Needs feature extraction or noise removal

TABLE II
DATASET 2: EPILEPTIC SEIZURE RECOGNITION.CSV

Property	Value	Description	Relevance
Samples	11,500	Total records (4:1 non-seizure:seizure ratio)	Balanced dataset size for robust evaluation
Features	175	Pre-extracted EEG features	Moderate dimensionality; manageable for classical ML
Class Balance	Imbalanced	About 20% minority class	Realistic clinical scenario
Feature Type	Extracted features	Statistical characteristics from EEG	Allows direct comparison: raw vs extracted

extremely high-dimensional. Together they show the interaction between preprocessing, imbalance treatment, and regularization.

III. PREPROCESSING PIPELINE DESIGN & RESULTS

Two preprocessing pipelines were tested to evaluate whether the order of preprocessing operations affects model performance.

TABLE III
DATASET 3: BONN EEG DATASET (BONN_EEG.CSV)

Property	Value	Description	Relevance
Samples	400	4 classes: F, O, S, Z (100 each)	Multi-class problem; tests generalization
Features	4,097	Raw EEG signal values	Extremely high-dimensional; stress tests dimensionality
Class Balance	Balanced	25% per class exactly	Isolates preprocessing effects without imbalance noise
Feature Type	Raw signals	Time-series window data	Requires aggressive feature extraction/reduction

A. Pipeline A: Normalization → Noise Removal → Feature Selection

Pipeline A emphasizes signal cleaning before dimensionality reduction.

- Robust scaling, resistant to outliers.
- Noise removal using outlier clipping based on the 1st–99th percentile range.
- Feature selection using SelectKBest with mutual information.

Rationale: preserve information-rich features by cleaning noise before selecting features.

B. Pipeline B: Feature Extraction → Scaling → PCA

Pipeline B extracts statistical features and then applies PCA for dimensionality reduction.

- Statistical feature extraction: mean, standard deviation, min, max, range.
- Scaling prior to dimensionality reduction.
- PCA reduction to 85% explained variance.

Rationale: capture interpretable signal summaries first, then remove correlated noise using PCA.

C. Preprocessing Results: Dataset-by-Dataset Comparison

TABLE IV
DATASET 1 (EEG_SCALED): PIPELINE A VS PIPELINE B

Metric	Pipeline A	Pipeline B	Difference	Winner
Accuracy	0.9439	0.8959	-0.048	Pipeline A
Recall	0.8049	0.5889	-0.216	Pipeline A
F1-score	0.7857	0.5909	-0.195	Pipeline A
PR-AUC	0.8153	0.5658	-0.250	Pipeline A

Interpretation: Pipeline A performs better on Dataset 1 because aggressive noise removal works well on highly imbalanced, noisy data.

Interpretation: Pipeline B dominates on Dataset 2. Statistical feature extraction captures meaningful EEG patterns better than raw feature selection, and PCA removes noise while preserving discriminative variance.

TABLE V
DATASET 2 (EPILEPTIC SEIZURE): PIPELINE A VS PIPELINE B

Metric	Pipeline A	Pipeline B	Difference	Winner
Accuracy	0.8017	0.9513	+0.150	Pipeline B
F1-score	0.0172	0.8713	+0.854	Pipeline B
PR-AUC	0.4886	0.9400	+0.451	Pipeline B

TABLE VI
DATASET 3 (BONN EEG, 4-CLASS): PIPELINE A VS PIPELINE B

Metric	Pipeline A	Pipeline B	Difference	Winner
Accuracy	0.4333	0.6333	+0.200	Pipeline B
F1-score	0.4074	0.6250	+0.218	Pipeline B
PR-AUC	0.3857	0.6723	+0.287	Pipeline B

Interpretation: In the highest-dimensional dataset, Pipeline B generalizes better because PCA provides stronger dimensionality reduction.

Key Finding

On 2 of 3 datasets, Pipeline B (Feature Extraction → Scaling → PCA) outperforms Pipeline A (Normalization → Noise Removal → Selection), with the greatest improvement on high-dimensional data.

IV. BASELINE LOGISTIC REGRESSION MODEL

A. Model Definition

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (1)$$

B. Why Logistic Regression?

- Coefficients are interpretable and indicate feature importance.
- Training is fast and suitable for comparing regularization methods.
- It provides a clear baseline to isolate the effect of preprocessing and imbalance handling.

TABLE VII
BASELINE PERFORMANCE ACROSS DATASETS

Dataset	Accuracy	F1-Score	Recall	PR-AUC
Dataset 1 (Binary)	0.816	0.156	0.083	0.431
Dataset 2 (Binary)	0.816	0.156	0.083	0.431
Dataset 3 (4-Class)	0.433	0.407	0.433	0.386
Average	0.688	0.240	0.200	0.416

Observation: baseline performs poorly on imbalanced datasets ($F1 < 0.2$) and moderately on the balanced multi-class setting. This confirms the importance of preprocessing and imbalance handling.

V. UNDERFITTING VS OVERFITTING ANALYSIS

A. Experimental Design

- Overfitting: Pipeline B (reduced features) with no regularization ($C = 1.0$).
- Good fit: Default regularization parameter ($C = 1$).
- Weak regularization: Pipeline A (high dimensional) with very weak regularization ($C = 10^6$).

B. Learning Curves: Train-Validation Gap Analysis

TABLE VIII
LEARNING CURVE ANALYSIS

Scenario	Training Acc	Validation Acc	Train-Val Gap	Diagnosis
Underfitting (Dataset 1)	0.510	0.511	0.0027	Model too weak
Good Fit (Dataset 1)	0.940	0.543	0.3971	Some overfitting
Overfitting (Dataset 1)	0.994	0.525	0.4690	Severe overfitting
Underfitting (Dataset 3)	0.293	0.511	0.2027	Under-trained
Overfitting (Dataset 3)	1.000	0.525	0.4750	Severely over-trained

C. Interpretation

- Underfitting: both training and validation accuracies are close to random; the model cannot learn the data well.
- Overfitting: training accuracy is near 100% but validation accuracy remains around 52–53%; the model memorizes the training set.
- Good fit: the model learns meaningful structure, but there is still a noticeable train-validation gap.

VI. REGULARIZATION STUDY: L1 VS L2 VS ELASTIC NET

A. Regularization Methods

The regularized logistic regression objective can be written as

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \|W\|^2. \quad (2)$$

- L1 (Lasso): encourages sparse models by forcing some coefficients to exactly zero.
- L2 (Ridge): shrinks coefficients proportionally and keeps them non-zero.
- Elastic Net: combines L1 and L2 penalties with $\alpha = 0.5$.

B. Dataset 1 Results

TABLE IX
DATASET 1 RESULTS: CLASS-WEIGHTED PIPELINE A

Regularization	Accuracy	Recall	F1-Score	PR-AUC
L1 (Lasso)	0.7303	0.8711	0.4521	0.7889
L2 (Ridge)	0.7294	0.8711	0.4513	0.7878
Elastic Net	0.7308	0.8711	0.4525	0.7886

TABLE X
DATASET 1 RESULTS: CLASS-WEIGHTED PIPELINE B

Regularization	Accuracy	Recall	F1-Score	PR-AUC
L1 (Lasso)	0.7855	0.8362	0.4990	0.6751
L2 (Ridge)	0.7846	0.8362	0.4979	0.6751
Elastic Net	0.7846	0.8362	0.4979	0.6749

TABLE XI
DATASET 2 AND 3 REGULARIZATION STRENGTH STUDY

C	L2 Acc	L2 F1	L1 Acc	L1 F1	Elastic Acc	Elastic F1	S
0.001	0.408	0.413	0.250	0.100	0.250	0.100	
0.01	0.442	0.446	0.250	0.100	0.250	0.100	
0.1	0.425	0.419	0.408	0.418	0.442	0.449	
1.0	0.433	0.407	0.425	0.431	0.433	0.434	

C. Dataset 2 and Dataset 3 Results

D. Sparsity Analysis (L1 Feature Selection)

Key Finding

Regularization has minimal impact on test performance. Across datasets and C values, L1, L2, and Elastic Net differ by less than about 1.5%. L1 is useful for sparse feature selection, while L2 is preferable for numerical stability.

VII. CLASS IMBALANCE HANDLING STRATEGIES

A. Techniques

- SMOTE: generates synthetic minority class samples.
- Undersampling: removes majority-class samples to balance the data.
- Class weighting: assigns a larger penalty to minority-class errors.

TABLE XII
L1 SPARSITY ANALYSIS

C Value	Non-Zero Features	Total Features	Sparsity (%)
0.001	0	2000	100.00
0.01	0	2000	100.00
0.1	71	2000	96.45
1.0	364	2000	81.80
10.0	477	2000	76.15
100.0	673	2000	66.35
1000.0	1048	2000	47.60

TABLE XIII
DATASET 1: IMBALANCE HANDLING IMPACT

Technique	Pipeline	Accuracy	F1-Score	PR-AUC
None (Baseline)	A	0.944	0.786	0.815
Undersampling	A	0.855	0.857	0.878
Class Weighting	A	0.932	0.754	0.814
None (Baseline)	B	0.896	0.591	0.566
Undersampling	B	0.834	0.836	0.821
Class Weighting	B	0.858	0.592	0.530

TABLE XIV
DATASET 2: IMBALANCE HANDLING METHODS

Technique	Accuracy	F1-Score	Precision	PR-AUC
SMOTE	0.696	0.362	0.313	0.413
Undersampling	0.679	0.334	0.287	0.393
Class Weight	0.706	0.348	0.309	0.398

TABLE XV
DATASET 3: BALANCED DATA

Technique	Accuracy	F1-Score	Precision
Baseline	0.433	0.407	0.550
SMOTE	0.433	0.407	0.550
Undersampling	0.433	0.407	0.550
Class Weight	0.433	0.407	0.550

B. Key Findings on Imbalance Handling

- Effectiveness depends on the preprocessing pipeline. For example, undersampling improves F1 much more on Pipeline B than Pipeline A.
- SMOTE performs best in the high-imbalance scenario of Dataset 2.
- Balanced datasets do not benefit from imbalance handling.
- Class weighting is practical because it avoids changing the data distribution.

VIII. COMPARATIVE ANALYSIS: ANSWERING KEY QUESTIONS

A. Question 1: Does Preprocessing Order Affect Results?

Answer: Yes, substantially.

TABLE XVI
PIPELINE COMPARISON ACROSS DATASETS

Dataset	Better Pipeline	Accuracy Delta	F1-Score Delta
Dataset 1 (Imbalanced, High-Dim)	Pipeline A	+4.8%	+19.5%
Dataset 2 (Imbalanced, Med-Dim)	Pipeline B	+15.0%	+85.4%
Dataset 3 (Balanced, Extreme-Dim)	Pipeline B	+20.0%	+21.8%

Pipeline B is the overall winner in 2 out of 3 datasets. The gains are larger on high-dimensional problems because PCA reduces dimensionality more effectively than greedy feature selection.

B. Question 2: Which Regularization Generalizes Best Across Datasets?

Answer: L1, L2, and Elastic Net are statistically equivalent across datasets.

Differences remain within the noise of train-test splits, so regularization choice should be made for interpretability or stability rather than predictive performance.

TABLE XVII
CROSS-DATASET REGULARIZATION COMPARISON

Dataset	L1 F1	L2 F1	Elastic F1	Max Gap	Conclusion
Dataset 1 (A)	0.4521	0.4513	0.4525	0.0012	Identical
Dataset 1 (B)	0.4990	0.4979	0.4979	0.0011	Identical
Dataset 2 ($C = 1.0$)	0.4306	0.4074	0.4341	0.0267	~Identical
Dataset 3 ($C = 1.0$)	0.4306	0.4074	0.4341	0.0267	~Identical

C. Question 3: Does Elastic Net Consistently Outperform L1/L2?

Answer: No, Elastic Net does not consistently outperform L1 or L2.

TABLE XVIII
HEAD-TO-HEAD COMPARISON OF REGULARIZATION METHODS

Metric	L1 Wins	L2 Wins	Elastic Wins
Accuracy (Datasets)	3/10	2/10	5/10
F1-Score (Datasets)	2/10	3/10	5/10
Sparsity Achieved	YES (100%)	NO	Partial (~50%)
Overall Winner	Marginal	Marginal	Marginal

Elastic Net is a compromise, but it gives up the sparsity of L1 and the simplicity of L2 without providing meaningful predictive gains.

D. Question 4: How Does Imbalance Handling Interact with Regularization?

Answer: Imbalance handling and regularization operate on orthogonal axes. Their effects are largely independent.

TABLE XIX
INTERACTION MATRIX (DATASET 1, PIPELINE A)

Imbalance Method	L1 Acc	L2 Acc	Elastic Acc	Reg Gap	Effect
None (Baseline)	0.944	0.944	0.944	0.000	No variance
Undersampling	0.855	0.855	0.855	0.000	No variance
Class Weight	0.730	0.729	0.731	0.002	Tiny variance

The dominant source of variation is imbalance handling; regularization choice contributes very little.

IX. SUMMARY OF KEY FINDINGS

- Preprocessing is the dominant factor.** Pipeline choice can improve F1 by up to 85%.
- Statistical feature extraction + PCA is better than raw noise removal.** Pipeline B performs better on most datasets, especially high-dimensional ones.
- Regularization type does not matter much for linear models.** L1, L2, and Elastic Net are statistically similar in predictive performance.
- Imbalance handling dominates performance.** The effect of class balancing is much larger than the effect of regularization.
- Underfitting and overfitting are distinct.** Learning curves clearly expose both behaviors.

- **Perfect balance does not require imbalance handling.** When classes are already balanced, resampling methods provide no benefit.

X. RECOMMENDATIONS FOR PRACTITIONERS

A. Prioritize Preprocessing

- Extract statistical features (mean, std, min, max, range) from raw signals.
- Normalize to zero mean and unit variance.
- Apply PCA to retain 85% explained variance.
- Prefer this pipeline over raw normalization plus greedy feature selection.

B. Handle Class Imbalance Appropriately

- Ratio $< 2:1$ $\rightarrow$ use class weighting.
- Ratio $2:1$ to $5:1$ $\rightarrow$ use SMOTE.
- Ratio $> 5:1$ $\rightarrow$ use undersampling.
- Balanced data $\rightarrow$ skip imbalance handling.

C. Choose Regularization by Use Case

- Need interpretability $\rightarrow$ use L1.
- Need stability $\rightarrow$ use L2.
- No preference $\rightarrow$ L2 is simpler; Elastic Net adds little value.

D. Monitor for Underfitting and Overfitting

- Plot learning curves with training and validation accuracy.
- Underfitting: both accuracies near random; reduce regularization or add features.
- Overfitting: training near 100%, validation near 50%; increase regularization or reduce features.
- Target a train-validation gap around 10–15% with both accuracies above 80%.

XI. CONCLUSION

This detailed study shows that seizure prediction performance is more dependent on the preprocessing method than on the imbalance-handling method, and only weakly dependent on the regularization method. Pipeline B, which combines statistical feature extraction and PCA, is the stronger general choice. The regularization methods L1, L2, and Elastic Net are statistically similar, so the choice should be based on interpretability and stability rather than accuracy. The largest gains come from aggressive preprocessing and thoughtful handling of class imbalance, not from tuning regularization alone.